Methodology card.

1. Project Information.

Title	Robotization and Wages in the US Manufacturing Industry
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GitHub repository	https://github.com/Homax06/robotization-wages-us-manufacturing

2. Type of research.

Quantitative (observational, panel-data study).

The research question asks whether two measurable quantities are statistically related: the intensity of robot use and the level of wages. Both are numeric indicators published by official statistical agencies, and the hypothesis is tested with regression analysis. No interviews or text analysis are needed, so a qualitative or mixed-methods component is not justified. Because the data are observed rather than produced by a controlled experiment, we can establish a statistical *association*, not proof of causation.

3. Research question and hypothesis.

Research question: Is there a statistically significant relationship between the level of robotization and the level of real wages in US manufacturing at the state level?

- **Null hypothesis (H_0):** $\beta_1 = 0$. There is no statistically significant relationship between the robot exposure share and real wages in US manufacturing.
- **Alternative hypothesis (H_1):** $\beta_1 \neq 0$. There is a statistically significant relationship between the robot exposure share and real wages.

Significance level: $\alpha = 0.05$.

Refinements:

1. H_1 is now two-sided. Earlier literature (e.g., Acemoglu & Restrepo, 2020) reports negative wage effects, so a directional “positive” hypothesis would not be justified.
2. The dependent variable is *real* (CPI-deflated) wages instead of nominal wages, so inflation does not contaminate the estimate.
3. The independent variable is the *percent of employees exposed to robots* (state level, US Census Bureau) instead of robots per 10,000 workers, because open state-level robot counts are not available.
4. The scope is the manufacturing sector as a whole (NAICS 31–33) instead of machinery manufacturing (NAICS 333). The Census Bureau publishes state-level robot data only for the whole manufacturing sector; for machinery (NAICS 333) the data exist only for the US as a whole.
5. The study period is 2018–2021, the years covered by the Census robotics data. The period is a configurable parameter.

4. Variable matrix.

Type	Variable	Operationalization and source
Independent	Robot exposure share, $ROBOT_{it}$	Percent of employees exposed to robots in manufacturing, state *i*, year *t*. Source: US Census Bureau, Annual Survey of Manufactures (ASM), robotics tables 2018–2021
Dependent	Log real wage, $\ln(W_{it})$	Natural log of average annual pay in NAICS 31–33 (BLS QCEW, private ownership), divided by CPI-U and expressed in constant dollars of the base year
Control covariates (in the model)	Industry size and structure	ln(employment) and ln(number of establishments) in NAICS 31–33 (BLS QCEW); year dummies (common shocks such as inflation and the business cycle)
Controlled (fixed) conditions	Experimental environment	Python and library versions (exact pins), random seed, dates and versions of downloaded data, CPI base year, industry code (NAICS 31–33), ownership (private), study period, sample rules (below) and standard-error type — all fixed in requirements.txt, Dockerfile and config.yaml

Sample rules. The panel covers 50 states and the District of Columbia for 2018–2021 (204 state-year cells). Cells withheld by the Census Bureau (codes D and S) are treated as missing, and the District of Columbia and Hawaii are excluded because most of their cells are withheld or reported as zero. Cells flagged “(s)” (standard error above 40% of the estimate) are kept in the main model and dropped in a robustness check.

5. Metrics and baseline.

Models.

- Baseline (M0): $\ln(W_{it}) = \alpha + \beta_1 \cdot ROBOT_{it} + \varepsilon_{it}$
- Main model (M1): $\ln(W_{it}) = \alpha + \beta_1 \cdot ROBOT_{it} + \gamma' X_{it} + \delta_t + \varepsilon_{it}$

where X_{it} are the covariates and δ_t are year effects. Standard errors are clustered by state.

Primary metric. The estimate b_1 of the coefficient β_1 from M1 with its 95% confidence interval and p-value ($t = b_1 / SE(b_1)$). H_0 is rejected if $p < 0.05$. Interpretation: +1 percentage point of exposure share is associated with roughly $100 \cdot b_1$ % change in real wages.

Secondary metrics. Adjusted R^2, RMSE, and the change in b_1 between M0 and M1.

Robustness checks. (a) Replace ROBOT with the Census “demeaned” exposure, which controls for differences in industry composition between states; (b) drop cells flagged “(s)”; (c) add state fixed effects.

Guardrail metrics.

Check	Criterion	Purpose
Variance inflation factor (VIF)	< 10	No harmful multicollinearity among regressors
Breusch–Pagan test	$p > 0.05$, otherwise rely on robust (clustered) errors	Detect heteroskedasticity
Residual normality (Q–Q plot, Jarque–Bera)	visual and formal check	Validity of t-tests
Sample coverage	at least 45 states per year; at most 10% of cells withheld	Enough data for inference
Pipeline reproducibility	two runs give identical results tables	Reproducibility of the analysis

A failed guardrail does not discard the result. It triggers a documented correction (for example, robust errors) or a stated limitation. The Durbin–Watson test is not used because the panel has only four years; clustering of standard errors by state handles within-state correlation instead.

Baseline. M0 is a simple regression of log real wage on robot exposure without any covariates. Comparing M0 with M1, on the same sample, shows how much the estimate changes once other factors are accounted for, which reveals omitted-variable bias.

6. Threats to validity and limitations

Threat	**Mitigation**
Omitted variables (e.g., productivity, union density, education)	Covariates, year effects, comparison of M0 and M1; state fixed effects as a robustness check
Industry composition (states specialize in more or less robot-intensive industries)	Robustness check with the Census “demeaned” exposure measure
Reverse causality (high wages may encourage automation)	Results are reported as associations, not causal effects
Short panel (4 years) and experimental data; some cells withheld	Disclosed limitation; coverage guardrail; flagged cells dropped in a robustness check
External validity: one sector, one country	Conclusions are limited to US manufacturing

7. Data source.

US Census Bureau — Annual Survey of Manufactures, Industrial Robotic Equipment 2018–2021 (file ASM Robotics.xlsx, sheets “Percent of… Geo” and “Percent of… Geo demean”).

US Bureau of Labor Statistics — Quarterly Census of Employment and Wages (QCEW), annual-average files (fields area_fips, own_code, industry_code, year, annual_avg_emplvl, annual_avg_estabs, avg_annual_pay); Consumer Price Index (CPI-U).